

W00 · Setup · 30 MINIT / MINUTES · Hari 1, 09:00 hingga 09:30

# Hands-On SQA-AI Assistant Workshop

## Set up your SQA-AI assistant

Sediakan pembantu AI SQA anda

Clone the course repository, check the PC, install opencode, start it in the lab folder with the free model and learn the five controls you will use all day.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

# Before you start / Sebelum mula

## YOU WILL / ANDA AKAN

1. Clone the repository and run setup\check-lab.ps1
2. Install opencode with npm and check the version
3. Start opencode in lab\ with the Big Pickle model
4. Learn Tab (Plan or Build), @file, /command, /models and !command

1. Klon repositori dan jalankan setup\check-lab.ps1
2. Pasang opencode dengan npm dan semak versi
3. Mulakan opencode dalam lab\ dengan model Big Pickle
4. Kenali Tab (Plan atau Build), @fail, /arahan, /models dan !arahan

## CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)

### Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

## OUTPUT

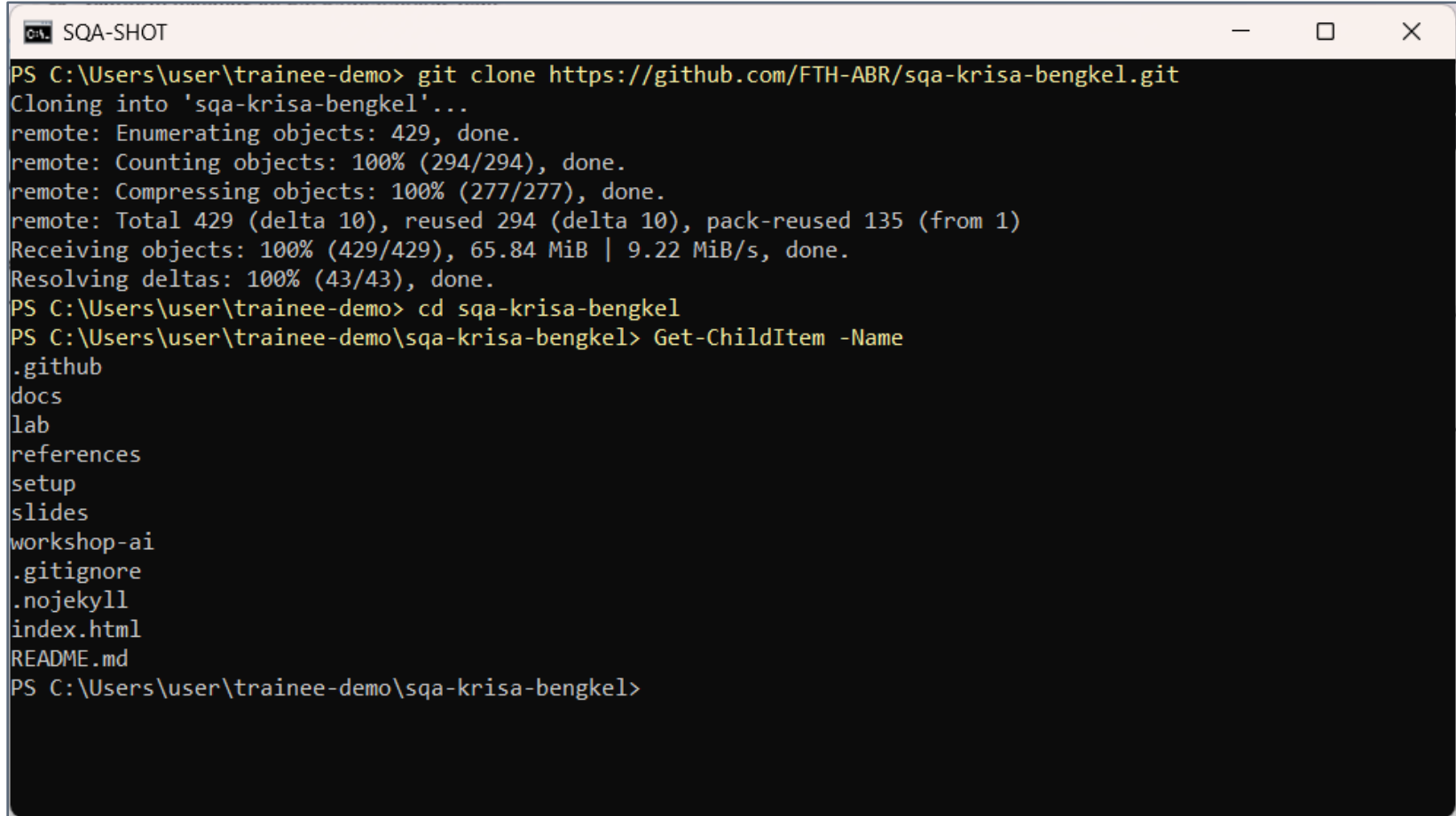
**PC sedia: check-lab.ps1 menunjukkan SEMUA OK**

## Open PowerShell, go to your home folder and clone the course repository.

Buka PowerShell, pergi ke folder rumah dan klon repositori kursus.

### TYPE IN POWERSHELL

```
cd $HOME
git clone https://github.com/FTH-ABR/sqa-krisa-bengkel.git
cd sqa-krisa-bengkel
```



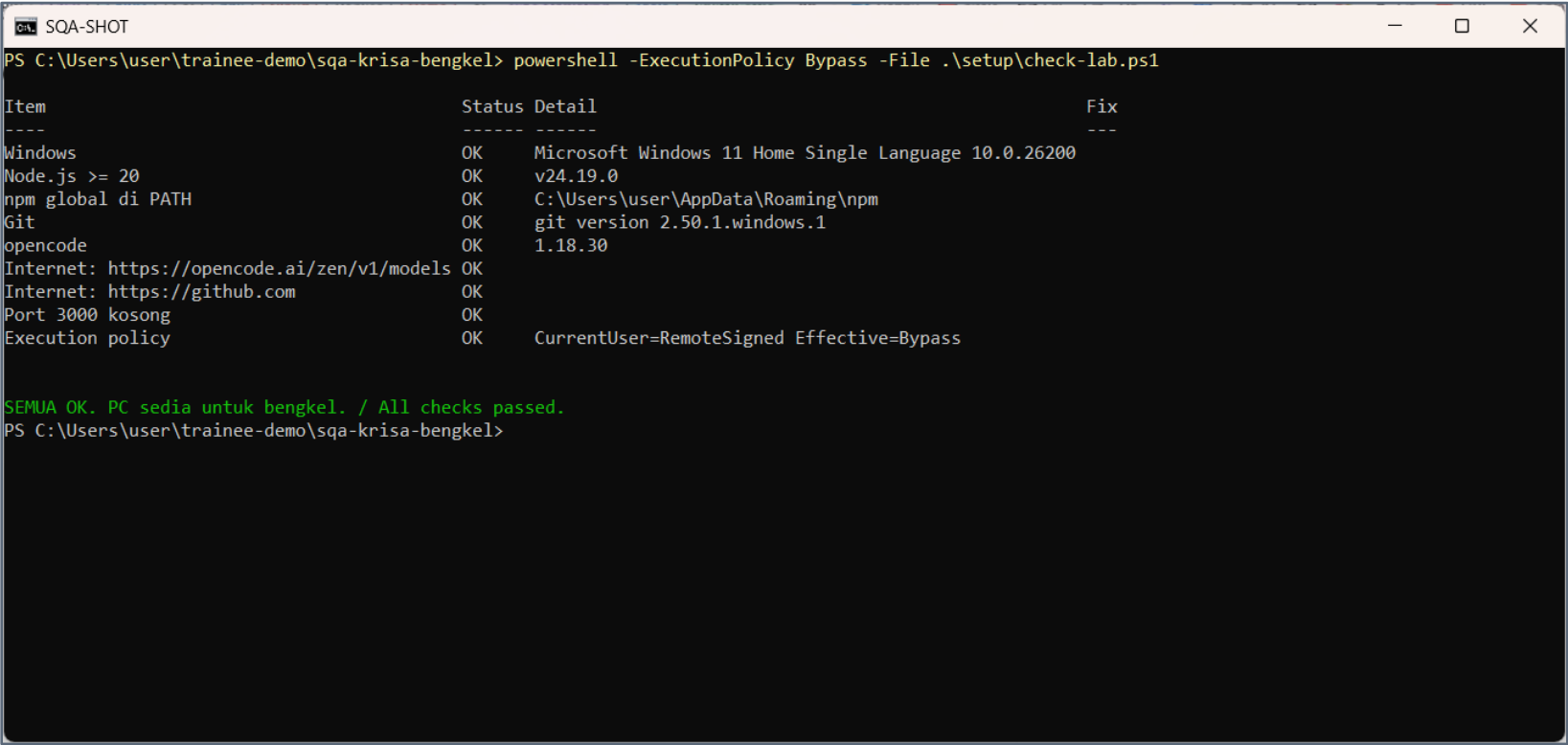
```
SQA-SHOT
PS C:\Users\user\trainee-demo> git clone https://github.com/FTH-ABR/sqa-krisa-bengkel.git
Cloning into 'sqa-krisa-bengkel'...
remote: Enumerating objects: 429, done.
remote: Counting objects: 100% (294/294), done.
remote: Compressing objects: 100% (277/277), done.
remote: Total 429 (delta 10), reused 294 (delta 10), pack-reused 135 (from 1)
Receiving objects: 100% (429/429), 65.84 MiB | 9.22 MiB/s, done.
Resolving deltas: 100% (43/43), done.
PS C:\Users\user\trainee-demo> cd sqa-krisa-bengkel
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> Get-ChildItem -Name
.github
docs
lab
references
setup
slides
workshop-ai
.gitignore
.nojekyll
index.html
README.md
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel>
```

Run the lab readiness check. Every line must say OK. Fix anything else using the Fix column.

Jalankan semakan kesediaan makmal. Setiap baris mesti OK. Betulkan yang lain menggunakan lajur Fix.

TYPE IN POWERSHELL

```
powershell -ExecutionPolicy Bypass -File
.\setup\check-lab.ps1
```

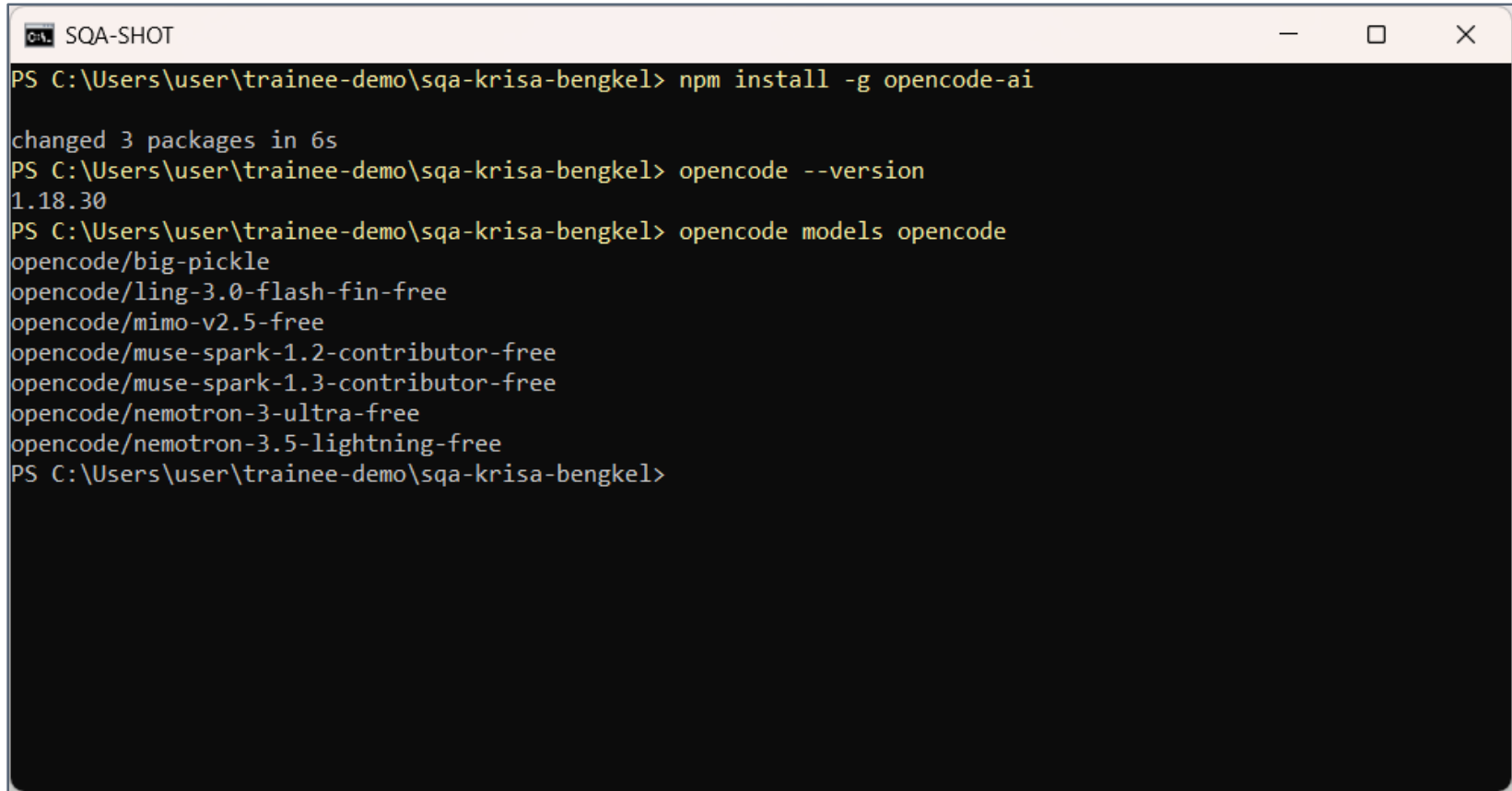


## Install opencode with npm, confirm the version and list the free OpenCode Zen models.

Pasang opencode dengan npm, sahkan versi dan senaraikan model percuma OpenCode Zen.

### TYPE IN POWERSHELL

```
npm install -g opencode-ai
opencode --version
opencode models opencode
```



```
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> npm install -g opencode-ai

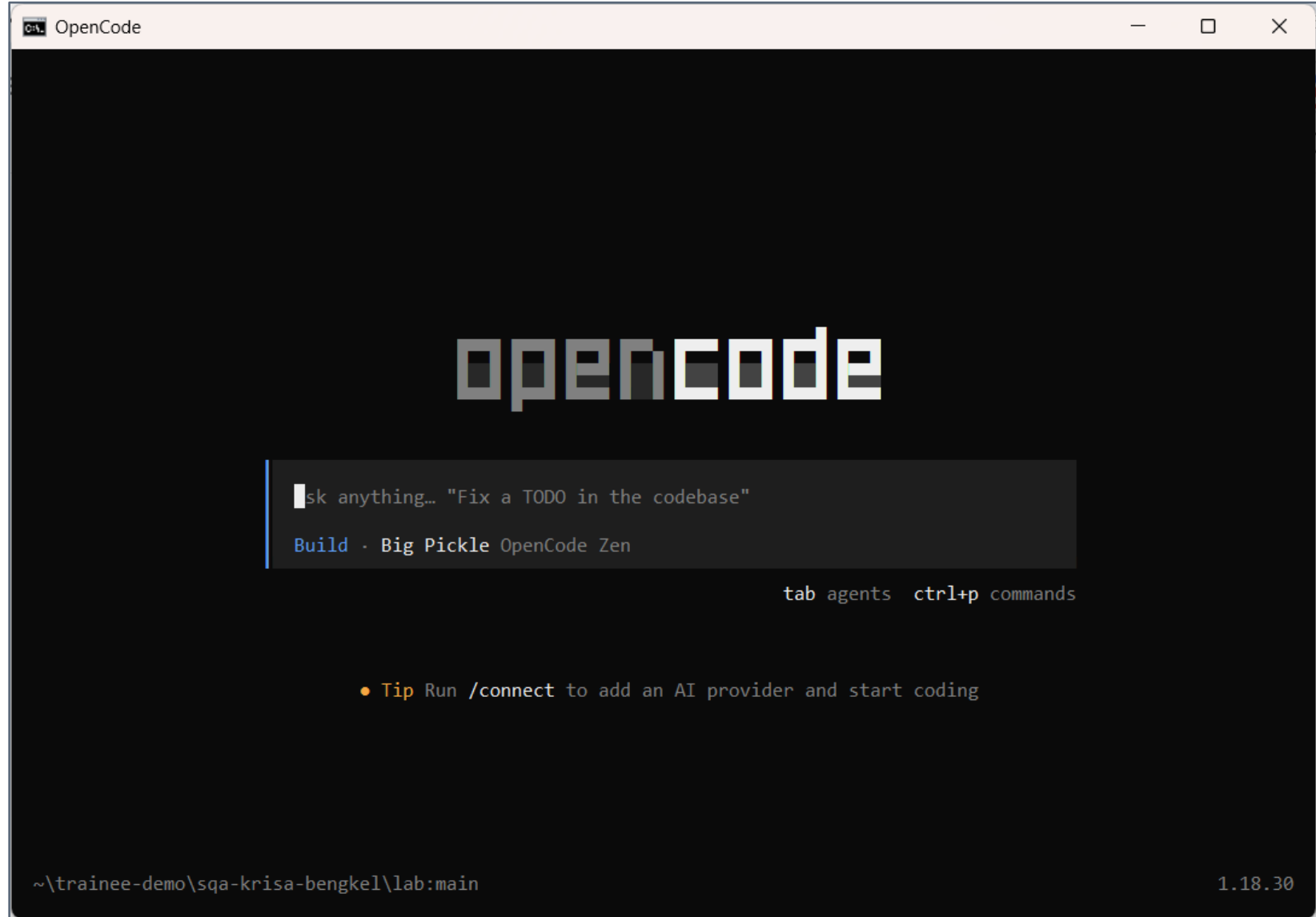
changed 3 packages in 6s
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> opencode --version
1.18.30
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> opencode models opencode
opencode/big-pickle
opencode/ling-3.0-flash-fin-free
opencode/mimo-v2.5-free
opencode/muse-spark-1.2-contributor-free
opencode/muse-spark-1.3-contributor-free
opencode/nemotron-3-ultra-free
opencode/nemotron-3.5-lightning-free
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel>
```

**Go into the lab folder and start opencode with the free Big Pickle model. No account or API key is needed.**

Masuk ke folder lab dan mulakan opencode dengan model percuma Big Pickle. Tiada akaun atau kunci API diperlukan.

**TYPE IN POWERSHELL**

```
cd lab
opencode -m opencode/big-pickle
```

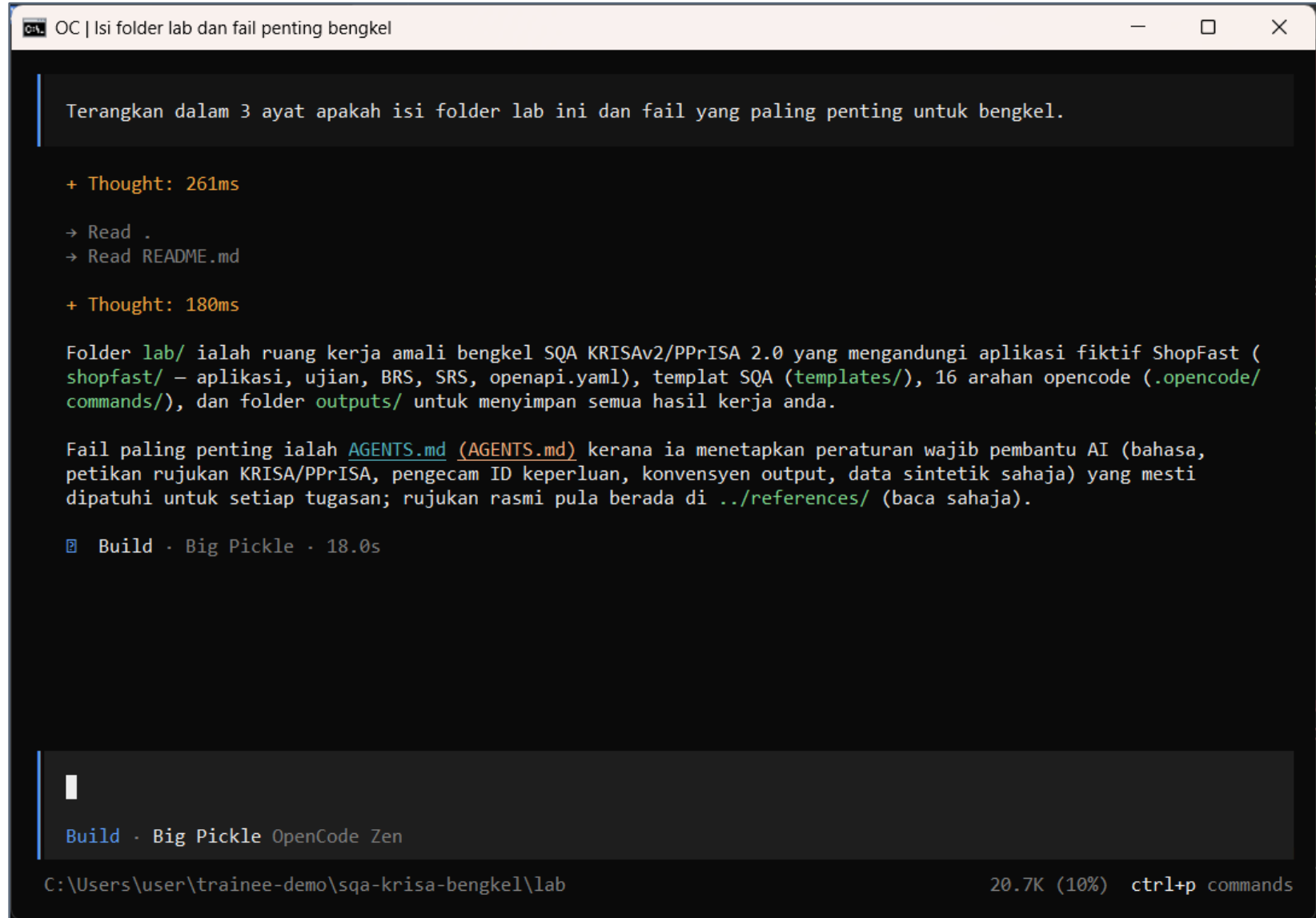


Type your first question and press Enter. The assistant reads the folder and AGENTS.md before answering.

Taip soalan pertama dan tekan Enter. Pembantu membaca folder dan AGENTS.md sebelum menjawab.

#### TYPE IN OPENCODE

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.



```
OC | Isi folder lab dan fail penting bengkel

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.

+ Thought: 261ms
→ Read .
→ Read README.md

+ Thought: 180ms

Folder lab/ ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (shopfast/ – aplikasi, ujian, BRS, SRS, openapi.yaml), templat SQA (templates/), 16 arahan opencode (.opencode/commands/), dan folder outputs/ untuk menyimpan semua hasil kerja anda.

Fail paling penting ialah AGENTS.md (AGENTS.md) kerana ia menetapkan peraturan wajib pembantu AI (bahasa, petikan rujukan KRISA/PPrISA, pengecam ID keperluan, konvensyen output, data sintetik sahaja) yang mesti dipatuhi untuk setiap tugas; rujukan rasmi pula berada di ../references/ (baca sahaja).

Build - Big Pickle - 18.0s

Build - Big Pickle OpenCode Zen

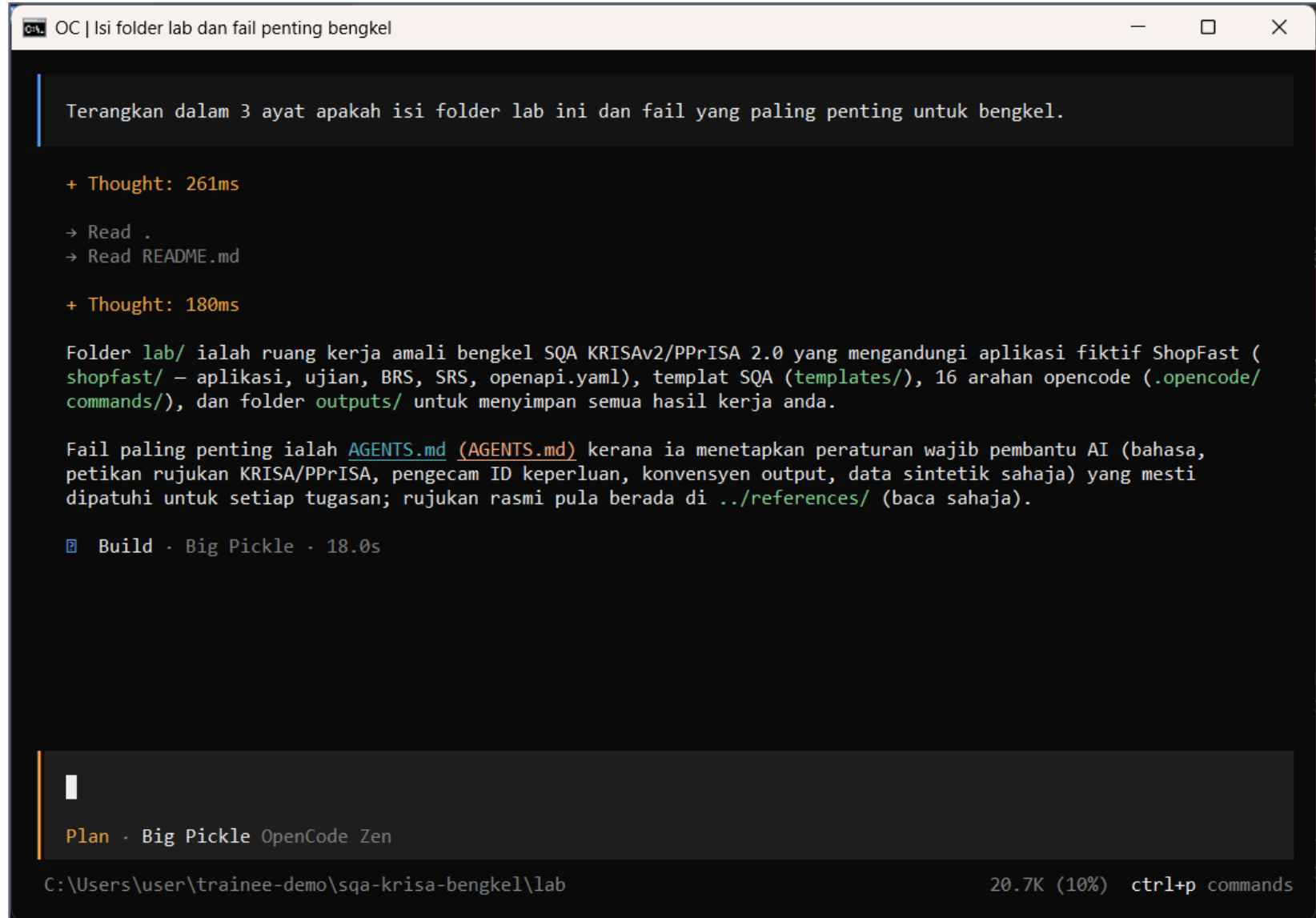
C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab 20.7K (10%) ctrl+p commands
```

**Press Tab to switch between Build (can edit files) and Plan (read only). Use Plan for reviews.**

Tekan Tab untuk bertukar antara Build (boleh ubah fail) dan Plan (baca sahaja). Guna Plan untuk semakan.

#### TYPE IN OPENCODE

Tab



OC | Isi folder lab dan fail penting bengkel

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.

+ Thought: 261ms

→ Read .  
→ Read README.md

+ Thought: 180ms

Folder `lab/` ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (`shopfast/` – aplikasi, ujian, BRS, SRS, `openapi.yaml`), templat SQA (`templates/`), 16 arahan opencode (`.opencode/commands/`), dan folder `outputs/` untuk menyimpan semua hasil kerja anda.

Fail paling penting ialah `AGENTS.md` (`AGENTS.md`) kerana ia menetapkan peraturan wajib pembantu AI (bahasa, petikan rujukan KRISA/PPrISA, pengecam ID keperluan, konvensyen output, data sintetik sahaja) yang mesti dipatuhi untuk setiap tugas; rujukan rasmi pula berada di `../references/` (baca sahaja).

Build · Big Pickle · 18.0s

Plan · Big Pickle OpenCode Zen

C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab 20.7K (10%) `ctrl+p` commands

Type @ and part of a file name to attach a file to your prompt. Pick it from the list.

Taip @ dan sebahagian nama fail untuk melampirkan fail pada prompt. Pilih daripada senarai.

#### TYPE IN OPENCODE

```
@shopfast/docs/inci
```

The screenshot shows the OpenCode IDE window titled "OC | Isi folder lab dan fail penting bengkel". The chat interface displays a user prompt: "Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel." The response includes a thought process, file reading actions, a detailed description of the lab folder structure, and the importance of the AGENTS.md file. A file list at the bottom highlights "shopfast/docs/incidents.md".

```
OC | Isi folder lab dan fail penting bengkel

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.

+ Thought: 261ms
→ Read .
→ Read README.md

+ Thought: 180ms

Folder lab/ ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (shopfast/ – aplikasi, ujian, BRS, SRS, openapi.yaml), templat SQA (templates/), 16 arahan opencode (.opencode/commands/), dan folder outputs/ untuk menyimpan semua hasil kerja anda.

Fail paling penting ialah AGENTS.md (AGENTS.md) kerana ia menetapkan peraturan wajib pembantu AI (bahasa, petikan rujukan KRISA/PPrISA, pengecam ID keperluan, konvensyen output, data sintetik sahaja) yang mesti dipatuhi untuk setiap tugas; rujukan rasmi pula berada di ../references/ (baca sahaja).

[ Build · Big Pickle · 18.0s ]

shopfast/docs/incidents.md

@shopfast/docs/inci

Plan · Big Pickle OpenCode Zen

C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab 20.7K (10%) ctrl+p commands
```

Type / to see commands. The course adds 16 SQA commands such as /review-srs and /testcases.

Taip / untuk melihat arahan. Kursus menambah 16 arahan SQA seperti /review-srs dan /testcases.

TYPE IN OPENCODE

```
/mod
```

OC | Isi folder lab dan fail penting bengkel

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.

+ Thought: 261ms

→ Read .  
→ Read README.md

+ Thought: 180ms

Folder lab/ ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (shopfast/ – aplikasi, ujian, BRS, SRS, openapi.yaml), templat SQA (templates/), 16 arahan opencode (.opencode/commands/), dan folder outputs/ untuk menyimpan semua hasil kerja anda.

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|                 |                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------|
| /models         | Switch model                                                                                         |
| /move           | Move to another project dir                                                                          |
| /change-request | M08 Isi Borang Permohonan Pindaan (PPrISA10) bagi perubahan pada ShopFast                            |
| /review-srs     | M02 Semak SRS ShopFast berbanding BRS dan senaraikan kecacatan keperluan (papar sahaja)              |
| /gen-api-tests  | M05 Jana ujian API automatik (Node.js node:test) daripada kes ujian ShopFast                         |
| /bug-report     | M06 Tulis laporan ralat (KRISAv2 6.3 dan PPrISA14) dengan severity Jadual 6.3                        |
| /test-summary   | M07 Sediakan laporan ringkasan ujian (D14 dan D16) dengan keputusan GO, NO-GO atau CONDITIONAL       |
| /audit          | M07 Audit kualiti ringkas (ISO 9001 dan CMMI lite) ke atas dokumen dan output bengkel (papar sahaja) |
| /risk-register  | M03 Bina daftar risiko ShopFast (PPrISA07) dengan skor 5x5 dan keutamaan ujian                       |
| /rtm            | M02 Bina Requirements Traceability Matrix (RTM) ShopFast dalam CSV                                   |

/mod

Plan · Big Pickle OpenCode Zen

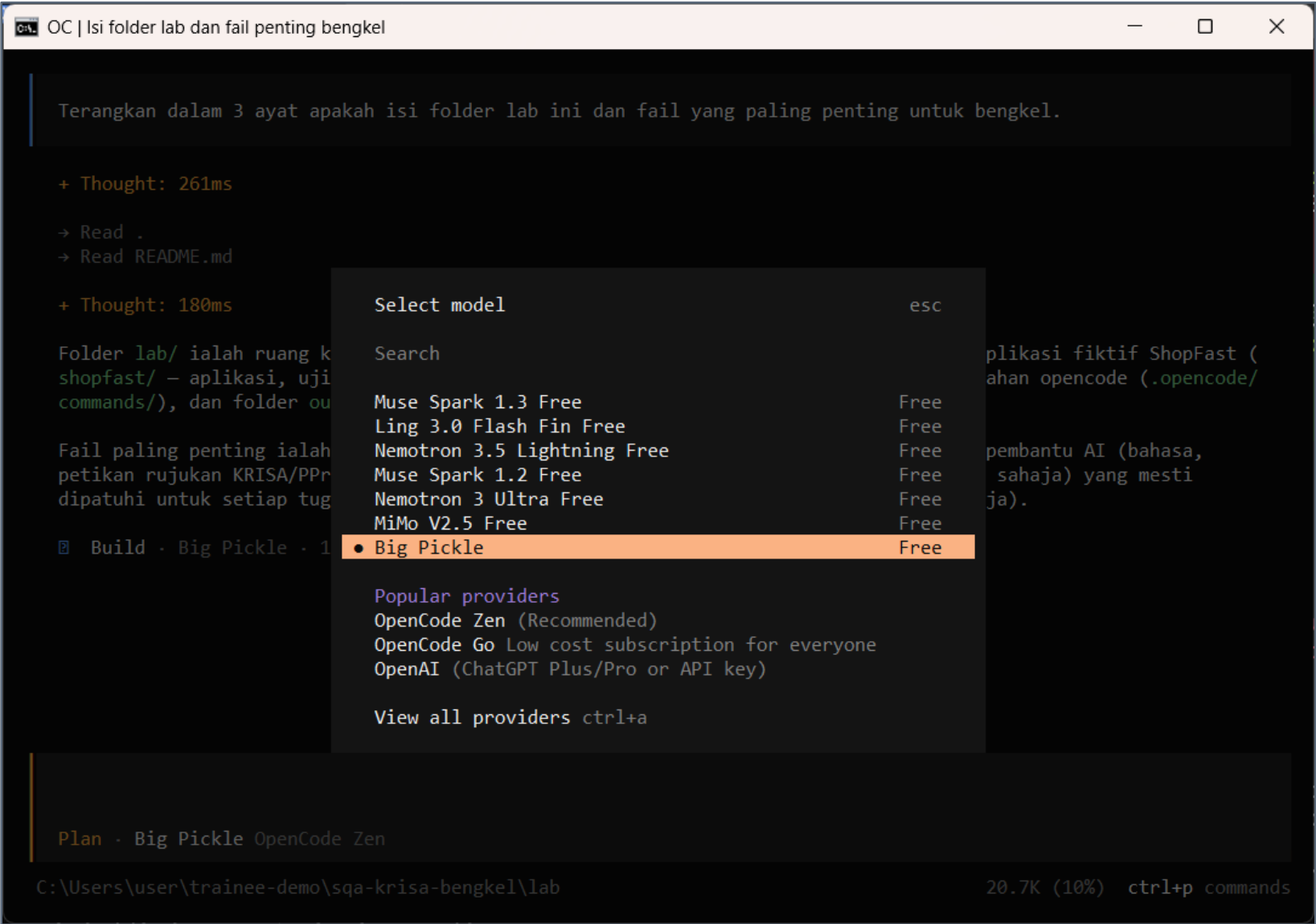
C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab20.7K (10%)ctrl+p commands

**/models shows the free models. Keep Big Pickle unless the trainer says otherwise.**

/models memaparkan model percuma. Kekalkan Big Pickle melainkan jurulatih mengarahkan lain.

TYPE IN OPENCODE

```
/models
```

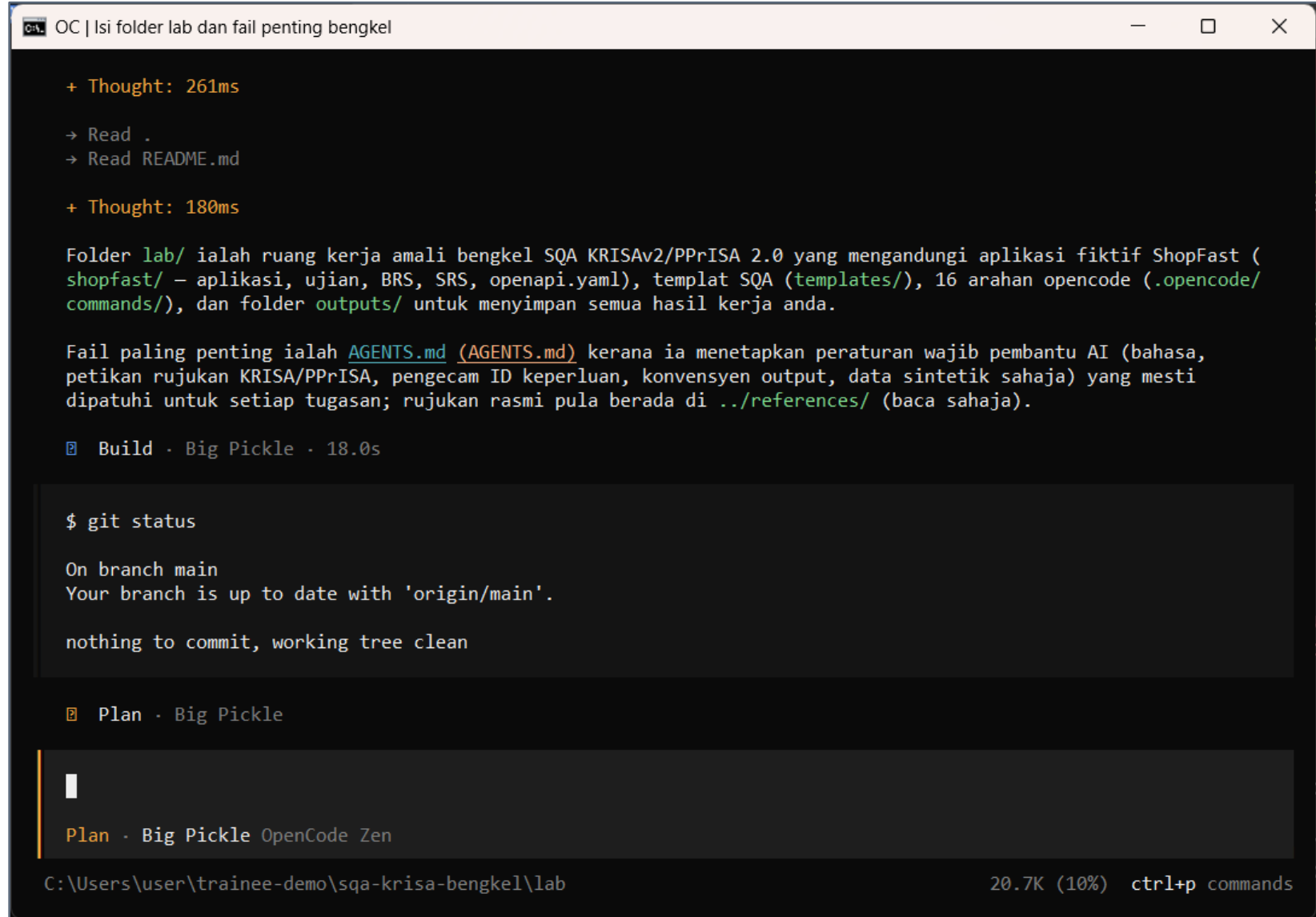


## Start a line with ! to run a terminal command without leaving opencode.

Mulakan baris dengan ! untuk menjalankan arahan terminal tanpa keluar dari opencode.

### TYPE IN OPENCODE

```
!git status
```



The screenshot shows the OpenCode IDE interface. The title bar reads "OC | Isi folder lab dan fail penting bengkel". The main editor area contains the following text:

```
+ Thought: 261ms
→ Read .
→ Read README.md
+ Thought: 180ms

Folder lab/ ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (shopfast/ – aplikasi, ujian, BRS, SRS, openapi.yaml), templat SQA (templates/), 16 arahan opencode (.opencode/commands/), dan folder outputs/ untuk menyimpan semua hasil kerja anda.

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```

Below the editor, there is a "Build" section showing "Build · Big Pickle · 18.0s". The terminal window displays the output of the command `$ git status`:

```
$ git status

On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Below the terminal, there is a "Plan" section showing "Plan · Big Pickle". At the bottom of the IDE, the status bar shows the file path `C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab`, the file size `20.7K (10%)`, and the command `ctrl+p commands`.

# Human check before you accept the output / Semakan manusia

## **check-lab.ps1 shows all OK**

check-lab.ps1 menunjukkan SEMUA OK

## **opencode --version prints 1.18.x**

opencode --version memaparkan 1.18.x

## **The status line shows Big Pickle, OpenCode Zen**

Baris status memaparkan Big Pickle, OpenCode Zen

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

# If something goes wrong / Jika ada masalah

## **opencode is not recognised**

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

## **Scripts are disabled on this system**

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

## **The model is slow or stops**

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

## **Permission required dialog**

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

## **File or command not found**

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

## **Answer uses Indonesian words or dashes**

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

## **Network or proxy error**

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting opencode

## **Port 3000 already in use**

`$env:PORT = 3001` before npm start, and use `http://localhost:3001`

# Abbreviations / Singkatan (1 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

| Abbreviation / Singkatan | Stands for / Maksud penuh                | In plain words / Maksud mudah                                                                                                                                                                                                                     |
|--------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI                       | Artificial Intelligence                  | Software that can answer questions and write text or code by learning patterns from large amounts of data.<br><i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i> |
| API                      | Application Programming Interface        | A doorway that lets one program ask another program for data or an action, without using a screen.<br><i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>                            |
| KRISA                    | Kejuruteraan Sistem Aplikasi Sektor Awam | The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition.<br><i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>                                          |
| MyIPO                    | Perbadanan Harta Intelek Malaysia        | Intellectual Property Corporation of Malaysia, the host of this training venue.<br><i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>                                                                                       |

# Abbreviations / Singkatan (2 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

| Abbreviation / Singkatan | Stands for / Maksud penuh                 | In plain words / Maksud mudah                                                                                                                                                                                              |
|--------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PC                       | Personal Computer                         | The lab computer you are working on.<br><i>Komputer peribadi, iaitu komputer makmal yang anda gunakan.</i>                                                                                                                 |
| PPriSA                   | Panduan Pengurusan Projek ICT Sektor Awam | The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates.<br><i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>                         |
| SQA                      | Software Quality Assurance                | All the planned work that gives confidence the software meets its quality requirements.<br><i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i> |
| TUI                      | Text User Interface                       | A full-screen program that runs inside a terminal window, like opencode.<br><i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>                              |

# Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)